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 Journal of Materials Research and Technology
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Original Article

Effects of rod size and fiber hybrid mode on the interface shear strength of carbon/glass fiber composite rods exposed to freezing-thawing and outdoor environments



Guijun Xian ^{a,b,c}, Rui Guo ^{a,b,c}, Chenggao Li ^{a,b,c,*}, Bin Hong ^d

^a Key Lab of Structures Dynamic Behavior and Control (Harbin Institute of Technology), Ministry of Education, Heilongjiang, Harbin, 150090, China

^b Key Lab of Smart Prevention and Mitigation of Civil Engineering Disasters of the Ministry of Industry and Information Technology, Harbin Institute of Technology, Harbin, 150090, China

^c School of Civil Engineering, Harbin Institute of Technology, Heilongjiang, Harbin, 150090, China

^d School of Transportation Engineering and Science, Harbin Institute of Technology, Heilongjiang, Harbin, 150090, China

ARTICLE INFO

Article history:

Received 12 May 2021

Accepted 21 August 2021

Available online 26 August 2021

Keywords:

Carbon/glass fiber reinforced polymer

Rod diameter

Fiber hybrid mode

Freezing-thawing and outdoor environments

Mechanical properties

Long-term life prediction

ABSTRACT

Fiber reinforced polymer composite rod has attracted wide attentions to replace the steel rod as bearing component in bridge cable and underground oil extraction. In the present paper, three kinds of carbon/glass fiber reinforced polymer hybrid rods were developed to reveal the effects of rod size (19 and 22 mm) and fiber hybrid mode (core-shell hybrid and random hybrid) on the interface shear strength exposed to freezing-thawing and outdoor environments. Water uptake, mechanical, thermal properties and density tests were conducted to obtain the long-term evolution. The fiber random hybrid mode brought about a slight increase of water uptake compared to the fiber core-shell hybrid mode. The effects of temperature and temperature-alternating on the water uptake acceleration factors were remarkable compared to the hydraulic pressure and salty concentration. Furthermore, the increase of rod diameter prolonged the saturated water uptake time up to 81.4–136.9%. The obvious degradation of interface shear strength was attributed to the hydrolysis and plasticization of the resin owing to the water uptake and the microcracks formation. After drying, the remarkable recovery up to 20.3% of interface shear strength can be observed. Based on the long-term life prediction, the effect of temperature-alternating on the degradation of interface shear strength was more remarkable compared to the constant temperature, leading to the lowest stable strength retention of 38.7%. Furthermore, the rods with the fiber random hybrid mode and smaller diameter possessed the superior corrosive resistances, which provide the long-term design guideline in the application of bridge cable and underground oil extraction.

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* Corresponding author.

E-mail address: lichenggao@hit.edu.cn (C. Li).

<https://doi.org/10.1016/j.jmrt.2021.08.088>

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1. Introduction

The premature performance degradation of the prestressed steel cable during the service will undoubtedly cause the reliability insufficiency and potential safety hazard of the bridge structures [1,2]. At the same time, the frequent replacements and maintenances of the steel cable will increase the cost of labors and materials, which will bring the obvious economy losses. In comparison, pultruded fiber reinforced polymer (FRP) composite rods possess the excellent performances, such as light weight, high strength and desirable corrosion resistance. These FRP rods have gained the wide attentions in some potential application fields to replace the traditional steel cable, such as bridge cable and underground oil extraction [3–6]. Through the reasonable replacement of the easy-corrosive steel rods with FRP rods, it is expected to achieve high performance and long life for the bridge engineering and oil extraction structures [7–9].

At present, the fiber type in the structural applications is mainly composed of carbon fiber, glass fiber, basalt fiber, aramid fiber and natural fiber. Furthermore, the mechanical properties and durability of single and hybrid composites including the above fibers are widely investigated [10–13]. Nurazzi et al. [14] summarized the research on the mechanical performance of hybrid natural fiber polymer composites for structural applications. They found the formation of displacement discontinuity surfaces contributed to the crack propagation of hybrid composites, leading to the decrease of mechanical properties. Furthermore, since the natural fiber-reinforced polymer composites do not provide the expected strength values, the basic studies related to strength, interface bonding and fracture mechanisms are suggested to be conducted to aid the future appropriate applications. Khalina et al. [15] investigated the effects of alkaline treatment and hybridization on the thermal properties of sugar palm yarn/glass fiber. The result obtained reveals that the thermal stability, char residue and decomposition temperature improved when the ratio of sugar palm and glass fiber increased up to 50/50 for both 30 and 40 wt.% of fiber loading. Owing to the demand of high beading capacity and long service life, FRPs used in the civil and oil extraction engineering mainly include carbon

fiber reinforced polymer (CFRP) composite and glass fiber reinforced polymer (GFRP) composite [16,17]. For glass fiber, the price is low, but the mechanical and fatigue properties are relatively poor [18]. In comparison, carbon fiber has high strength and modulus, excellent fatigue and corrosion resistances, however, the high price undoubtedly limits its application field [19,20]. To balance the material cost and performance, carbon/glass hybrid fiber reinforced polymer (HFRP) composite rod can take advantage of high mechanical and corrosion resistances of carbon fiber and the low cost and high deformation of glass fiber to achieve high cost performance [21–24]. Furthermore, the potential structural applications of hybrid rods include the bridge cable, prestressed tendon of concrete structures, bearing components in underground oil extraction and ground anchor [25–27]. Lammlein et al. [28] applied the combination of low clinker high performance concrete and ultra-high modulus CFRP tendons for prestressed structural elements. They found the CFRP prestressed tendons showed both higher prestressed losses of around 40% and, on average, a nearly doubled prestressed transfer length. Zhang et al. [29] conducted the simulation of ultralong stroke oil pumping unit used carbon fiber continuous sucker rod. The equivalent dynamics model of oil pumping unit and the longitude vibration model of the carbon fiber sucker rod were developed through considering the variation of rotating speed of electromotor. At last, it is proved the carbon fiber sucker rod has a bright application future. SENTRY et al. [30] studied the durability of carbon fiber reinforced polymer strands in ground anchors. Experiments were conducted in an unstressed state to analyze the absorption and physical effects as well as the ultimate tensile performance over a range of accelerated conditions and aggressive solutions. It can be found that the levels of corrosion protection systems required for CFRP anchor systems were critically assessed. The results highlighted the importance of identifying the CFRP strand resin matrix before it can be implemented in a ground anchor system. When the hybrid rods are applied for the above structures, the long-term property evolution is significant during exposure to the complicated service environment, such as the temperature, temperature-alternating, humidity, hydraulic pressure and static/dynamic loading [31]. Under the service condition, the hybrid rods may

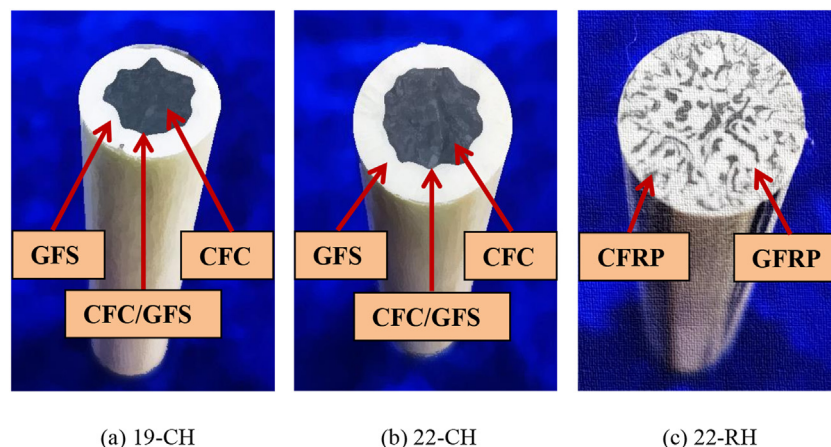


Fig. 1 – Cross-section schematic diagram of three kinds of hybrid rods.

Table 1 – The exposure conditions of three kinds of hybrid rods.

Exposure place	Exposure time (days)	Exposure temperature	Exposure medium
Freezing-thawing cycle chamber (laboratory environment)	0	–25 ~ 40 °C	Distilled water
	45		
	90		
	360		
Outdoor environment in Harbin (actual environment)	0	Actual outdoor temperature ¹	Air
	45		
	90		
	360		

Note: the data of actual outdoor temperature¹ was from China Meteorological Observatory (www.nmc.cn) for 2020.

undergo the resin hydrolysis, plasticization, and interface debonding of fiber/resin, leading to the degradation of mechanical properties [17,32,33]. Therefore, the investigation of long-term properties of hybrid rods exposed in service environments is urgent to put forward the evaluation method of the durability.

So far, the relevant investigations on the long-term properties of FRPs aimed at the single type fiber reinforced polymer composite, such as CFRP and GFRP [34–37]. The studies showed the degradation of mechanical properties was mainly attributed to the interaction results among the mechanical properties of fiber, physical and chemical evolution of resin, bonding properties of fiber/resin interface and service environment [38]. Furthermore, the diffusion of water molecule in the FRP was the main factor to contribute the degradation of mechanical properties [39]. Wu et al. [40] found that the interface degradation of fiber/resin was the main factor to cause the degradation of mechanical property, and loading has the accelerating effect. Lu et al. [41] concluded that degradation of resin can easily lead to the interface debonding of fiber/resin. Rutkowska et al. [42] found that the degradation of tensile properties was related to the crosslinking degree of resin. Hong et al. [33,43] found that the degradation of the fiber/resin interface was the reason of load transfer failure between the fiber and resin. However, the degradation

mechanism of single type fiber reinforced polymer composite exposed to the service environment may be not appropriate for the hybrid fiber composite when considering the synergistic effect between the fibers.

Currently, the investigations on the long-term evolution of hybrid rod were limited and concentrated on the hygro-thermal exposure at constant temperature. Li et al. [24,44] concluded that the exposed temperatures and hydraulic pressure accelerated the water absorption and diffusion through accelerating the water molecules filling into the initial defects of hybrid rods. Immersion in distilled and salty solutions deteriorated the interface shear strength of hybrid rod. Furthermore, the increase of immersed temperature led to significant degradation compare to hydraulic pressure and higher salty solution concentration. Barjasteh and Nutt [45] as well as Tsai et al. [46] found the temporary moisture barrier in the CFC/GFS interface contributed to the decrease of shear strength and glass transition temperature (T_g) of hybrid rods. Xian et al. [23,47] found that multiple hydrogen bonds are formed between the water molecules with the chain structures of resin, leading to the hydrolysis, plasticization of resin and interface debonding of fiber/resin. Furthermore, higher exposed temperature and hydraulic pressure aggravated the degradation of interface shear strength. Pan et al. [13] studied the durability of GFRP bars and carbon/glass hybrid fiber reinforced polymer (HFRP) bars exposed in alkaline solution. The results indicated that the remaining interlaminar shear strengths of HFRP bars were larger than that of GFRP bars. Furthermore, the deterioration of HFRP bars aged in alkaline solution was found occurring at the CFRP/CFRP interface. The CFRP layer enhanced the durability of GFRP bars in alkaline environment due to the delayed water and OH^- diffusion in CFRP and the better alkaline resistance of CFRP. The existing investigation mainly aimed at the long-term property evolution of rod with the fiber shell–core hybrid mode exposed in the constant temperature. However, the fiber hybrid mode, rod size and temperature-alternating were three significant factors when considering the material design and application field.

In the present study, three kinds of carbon/glass fiber reinforced polymer hybrid rods were developed through the pultrusion technology. The effects of rod diameter and fiber hybrid mode on the water uptake behavior and mechanical properties were investigated. The above hybrid rods were exposed in laboratorial freezing-thawing (FT) and actual outdoor environments (OE) as long as 360 days. The water uptake

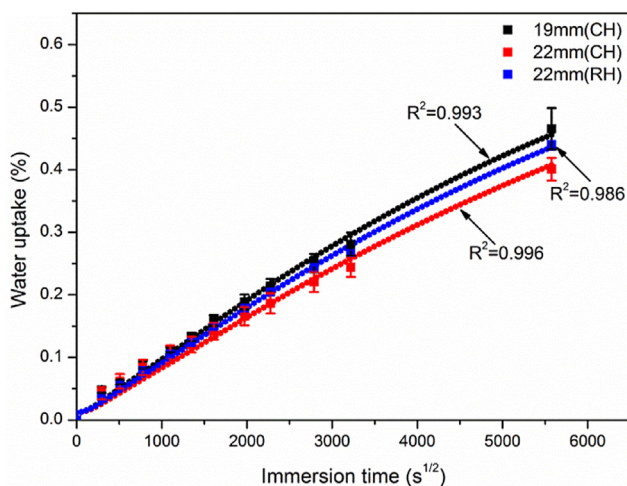


Fig. 2 – Water uptake versus the square root of time for hybrid rods exposed to the freezing-thawing environment.

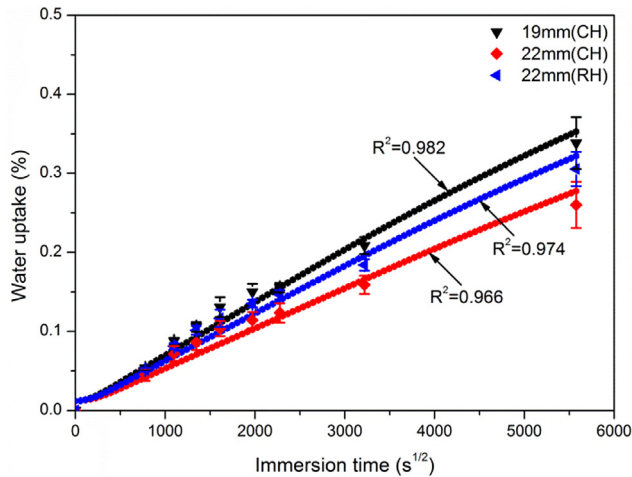


Fig. 3 – Water uptake versus the square root of time for hybrid rods exposed to outdoor environment.

and mechanical properties tests were conducted to obtain the long-term evolution. The thermal property and density tests were characterized to reveal the degradation mechanism. Based on the degradation mechanism, the long-term life prediction of hybrid rods was obtained to provide the design guideline of engineering applications.

2. Materials and methods

2.1. Raw materials

Through the pultrusion process, three kinds of carbon/glass fiber reinforced epoxy hybrid rods were developed to investigate the size effect and fiber hybrid mode, including the fiber core–shell hybrid (CH) mode with the diameter of 19 and 22 mm and the fiber random hybrid (RH) mode with the diameter of 22 mm. For convenience, the above three

types of hybrid rods were named as 19-CH, 22-CH and 22-RH, respectively, as shown in Fig. 1. As shown, the rods of 19-CH and 22-CH included the glass fiber reinforced epoxy shell (GFS) with a thickness of 3.5/4.0 mm and carbon fiber reinforced epoxy core (CFC) with the diameter of 12.0/14.0 mm. Meanwhile, the obvious irregular CFC/GFS interface can be observed for the fiber shell–core hybrid structure. By comparison, the rod of 22-RH realized the random distribution of carbon and glass fibers through the yarn spreading technology in the process of pultrusion. Plenty of mesoscale CFC/GFS interfaces existed in the fiber random hybrid structure. For the above three kinds of hybrid rods, the total volume contents of fibers were appropriately 70%, and the volume ratio of CFC and GFS was close to 2:3. The diameters of carbon fiber (Shanghai Petrochemical Co., Ltd, Shanghai, China) and glass fiber (OC 2400tex, Owens corning, America) were appropriately 7 and 20 μm , respectively. The matrix adopted the bisphenol-A epoxy matrix (Airstone™ 1122E, Blue Cube Chemicals Company, Zhangjiagang, China). It was worth mentioning that three kinds of hybrid rods were heated before the exposure in an oven at 120 °C for 48 h to guarantee the sufficient post-curing of resin matrix.

2.2. Methods

2.2.1. Exposure conditions

Two exposure conditions were applied for the present three types of hybrid rods as shown in Table 1, including the laboratory freezing-thawing (FT) cycle and actual outdoor environments (OE). As shown, the temperature-alternating of –25 to 40 °C for each freezing-thawing cycle was achieved in the freezing-thawing cycle chamber. The durations of freezing cycle and thawing cycle were 12 h, respectively, and the exposure medium was the distilled water. The outdoor exposure was the actual outdoor environment in Harbin (Heilongjiang, China), and the exposure medium was air. For the above two environments, the intermittent exposure time

Table 2 – Effects of temperature, temperature-alternating, hydraulic pressure, salty concentration, rod diameter and fiber hybrid mode on the average radial diffusivity coefficient of hybrid rods.

Factors	$D_r (\times 10^{-13}, \text{m}^2/\text{s})$			
Temperature (CH-DW) [44,46]	20/40/60 °C (0 MPa-19 mm)	90 °C/0 MPa (9.53 mm)	–25–40 °C/0 MPa (19 mm)	OE/0 MPa (19 mm)
	1.26/2.30/8.14	27.0	5.32	2.69
Hydraulic pressure (CH-DW-19 mm) [44]	60 °C/0 MPa	60 °C/20 MPa	/	/
	8.14	11.17		
Salty concentration (CH-19 mm) [24]	60 °C/0 MPa (DW)	60 °C/0 MPa (LSSC)	60 °C/20 MPa (LSSC/HSSC)	80 °C/20 MPa (LSSC)
	8.14	7.49	9.95/12.02	13.96
Diameter (CH-DW)	19/22 mm	19/22 mm (OE/0 MPa)	/	/
	(–25 to 40 °C/0 MPa)			
	5.32/3.89	2.69/1.53		
Hybrid mode (DW-22 mm)	CH (–25 to 40 °C/0 MPa)	RH (–25 to 40 °C/0 MPa)	CH (OE/0 MPa)	RH (OE/0 MPa)
	3.89	4.68	1.53	2.16
Core-shell (CH-7.75 mm) [45]	Hybrid rod (60 °C/85% R. H.)	Core-carbon (60 °C/85% R. H.)	Shell-glass (60 °C/85% R. H.)	/
	7.50	8.30	6.30	/
Core-shell (CH-19 mm) [44]	Hybrid rod (20/40/60 °C)	Core-carbon (20/40/60 °C)	Shell-glass (20/40/60 °C)	/
	1.26/2.30/8.14	1.39/2.55/9.01	1.06/1.93/6.84	/

Note: DW is the distilled water, LSSC is low salty solution concentration, HSSC is high salty solution concentration, R. H. is the humidity.

Table 3 – Maximum water uptake of hybrid rods exposed in laboratory and actual service environments (%).

19-CH (actual service environment) [9]				22-CH (actual service environment) [47]			
25 °C (211 days)	25 °C (606 days)	90 °C (211 days)	90 °C (606 days)	25 °C (255 days)	25 °C (553 days)	90 °C (255 days)	90 °C (553 days)
0.198	0.319	0.518	0.596	0.26	0.31	0.58	0.59
19/22 (laboratory freezing-thawing environment)				19/22 (actual outdoor environment)	19-CH (laboratory constant temperature immersion) [44]		
19-CH (365 days)	22-CH (365 days)	22-RH (365 days)	19-CH (365 days)	22-CH (365 days)	22-RH (365 days)	20/40/60/60 °C –20MPa (365days)	
0.47	0.40	0.44	0.34	0.26	0.31	0.24/0.32/0.51/0.54	

Note: the effects of immersed temperature, immersed time, rod diameter and fiber hybrid mode on the water uptake of hybrid rods were summarized, and the effects of immersion medium was ignored.

was 0, 45, 90 and 360 days. After a fixed exposure time, the mechanical and thermal performance tests of hybrid rods were conducted.

2.2.2. Water uptake tests

According to ASTM D5229, the water uptake tests of three kinds of hybrid rods exposed to the laboratorial and actual environments were conducted at room temperature (RT). The lengths of five hybrid rods were cut into 100 mm for each condition and the both ends were capped with an epoxy to simulate the radial diffusion of water molecule in the rod. To obtain the data of water uptake accurately, the test interval time was 1, 3, 7, 14, 21, 30, 45, 60, 90, 120 and 360 days. Before the exposure, the hybrid rods were firstly dried in an oven at 60 °C for 2 days to remove the surface moisture and low-molecular-weight species. After the periodic exposure, the mass variations of hybrid rods were obtained through the electronic balance with an accuracy of 0.1 mg. Based on the following equation, the water uptake of hybrid rods were obtained:

$$W(\%) = \frac{W_t - W_0}{W_0} \times 100\% \quad (1)$$

where W (%) is the water uptake of hybrid rod at t , W_t is the wet mass at time t , and W_0 is the control mass.

2.2.3. Drying tests

To analyze the existence form and content of water molecule in the hybrid rods, the drying tests after the above water absorption were conducted at RT. Two kinds of drying temperatures were selected to obtain the free water molecule and bound water molecule, and the corresponding temperatures were 60 and 90 °C, respectively. The content of residual bound water molecule can be further calculated. Based on the previous study [9], the longest drying time at 60 °C was 30 days to obtain the stable free water molecule and the drying time at 90 °C was 10 days to obtain the bound water. After the periodical drying, the hybrid rods were removed from the oven and measured the mass and density variations. It should be mentioned that the hybrid rods after the exposure of 360 days were also dried at 60 °C for 48 h to test the interface shear strength and glass transition temperature.

2.2.4. Interface shear strength (ISS)

The test method of interface shear strength (ISS) of CFC/GFS for three kinds of hybrid rods was based on the previous study [48]. The interface shear strength test device included the pressure lever, fixed roof and bearing pedestal. The diameters of pressure lever were 12 mm for 19-CH and 14 mm for 22-CH and 22-RH, respectively. After the hybrid rods were pressed to the bottom of bearing pedestal, the fixed roof and pressure lever were assembled and put on the base of universal tensile machine. A compressive force was applied on the pressure lever until the interface shear failure of hybrid rod. Five samples were prepared to test for each condition at RT and the crosshead displacement rate of universal tensile machine (DHY-10080, Shanghai, China) was set to 1.0 mm/min.

2.2.5. Dynamic mechanical analysis (DMA)

To analyze the degradation of hybrid rods during the exposure, the dynamic mechanical analysis were conducted through the DMA-Q800 (TA Instruments, America) according to the ASTM E1640-99. Two samples with the size of $40 \times 5 \times 2$ mm were prepared for the CFC and GFS for 19-CH and 22-CH, HFRP for 22-RH. The type of clamp was the single cantilever beam, and the load frequency and heating rate were 1 Hz and 5 °C/min from room temperature to 250 °C, respectively.

2.2.6. Density tests

The density tests of hybrid rods during the exposure and after drying adopted the drainage method. Five samples of 19-CH, 22-CH and 22-RH were prepared and put into a measuring cylinder filled with the distilled water at RT. The density was calculated according to the mass and volume variations of the samples. Five repeated tests were conducted for each condition to obtain the average density.

3. Results and discussion

3.1. Water absorption behavior

Fig. 2 shows the water uptake versus the time of three kinds of hybrid rods exposed to the freezing-thawing environment.

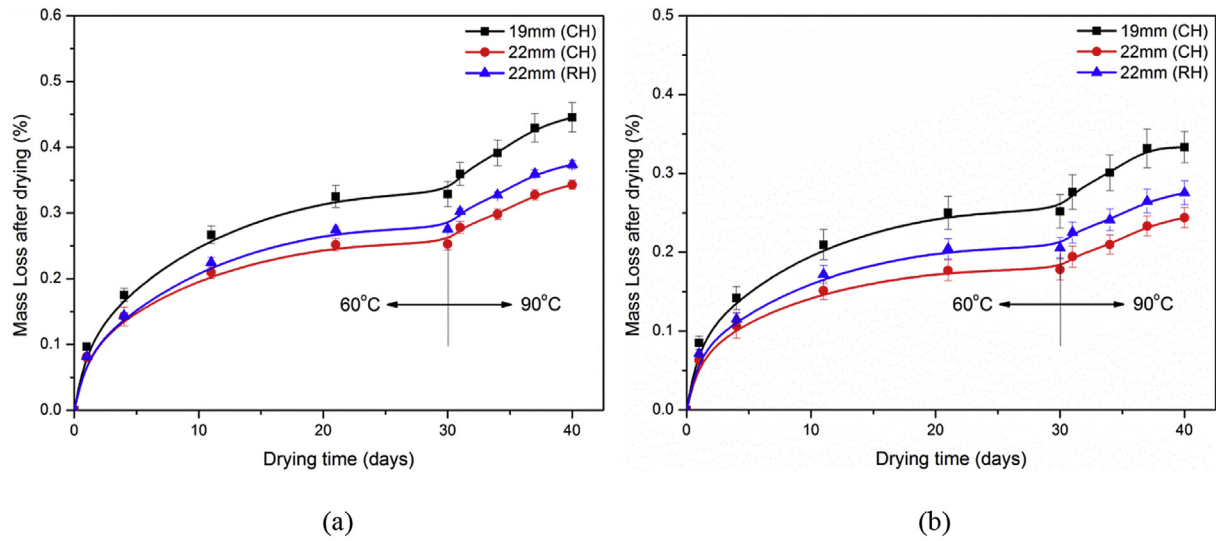


Fig. 4 – Mass loss of hybrid rods with the drying time of (a) freezing-thawing environment and (b) outdoor environment.

Furthermore, the water uptake in arbitrary time can be described by the Fick law as following:

$$\frac{M_t}{M_\infty} = 1 - \sum_{n=1}^{\infty} \frac{4}{R^2 \alpha_n^2} \exp(-D_r \alpha_n^2 t) \quad (2)$$

where M_t is the water uptake gain in time t , M_∞ is the saturation water uptake, R is the radius of hybrid rods, α_n is the n^{th} root of the zero-order Bessel function, and D_r is the average radial diffusivity coefficient. The higher degree of fitting ($R^2 > 0.98$) indicated the water uptake behaviors of three kinds of hybrid rods were accord with the classic Fick diffusion law.

As seen, the water uptake of hybrid rods shows the nonlinear variation with the immersion time. The increase rate in the initial stage was fast and gradually tended to be saturated. The maximum water uptakes at the longest immersion time of 360 days were 0.47, 0.40 and 0.44% for 19-CH, 22-CH and 22-RH, respectively. It can be found that the total water uptake of 19-CH was higher than that of 22-CH, this was because the larger diameter increased the interface areas of CFC/GFS, which remarkably increased the water absorption barrier effect through prolonging the diffusion paths [44]. In addition, the fiber random hybrid mode (22-RH) brought about a higher water uptake during the immersion. Possible reason

for this phenomenon was because the more interfacial areas of CFC/GFS for the fiber random hybrid structure provided more storage spaces for water molecule.

Fig. 3 shows the water uptake versus the immersion time for hybrid rods exposed to outdoor environment. Similar variation trends of water uptake behavior can be observed between the freezing-thawing and outdoor environments. In comparison, the maximum water uptake was lower for the hybrid rods exposed to the outdoor environments, and they were 0.34, 0.26 and 0.31% for 19-CH, 22-CH and 22-RH, respectively. This was because the maximum temperature and temperature-alternating effect of freezing-thawing were harsher than that of outdoor environment, which accelerated the diffusion of water molecule in the hybrid rods.

To analyze the water uptake behavior, Table 2 summarized the effects of temperature, temperature-alternating, hydraulic pressure, salty concentration, rod diameter and fiber hybrid mode on the average radial diffusivity coefficient of hybrid rods. When ignoring the effect of rod diameter, it can be seen that the diffusivity coefficient of hybrid rods increased obviously with the increase of immersed temperature from 20 to 90 °C. Furthermore, the diffusivity coefficients of the present immersion conditions (FT and OE) were between 40 and 60 °C,

Table 4 – The content of free and bound water for the present three kinds of hybrid rods (%).

Water type	Freezing-thawing environment			Outdoor environment		
	19-CH	22-CH	22-RH	19-CH	22-CH	22-RH
Free water (60 °C)	0.329 (70.7%)	0.253 (63.0%)	0.275 (62.7%)	0.252 (74.5%)	0.178 (68.4%)	0.206 (67.2%)
Bound water (90 °C)	0.116	0.090	0.098	0.081	0.066	0.070
Residual bound water	0.020	0.058	0.065	0.005	0.016	0.031

Note: the free water content was obtained through the balanced mass loss at 60 °C for 30 days. The bound water at 90 °C was obtained through the difference value of mass loss between 90 °C for 10 days and 60 °C for 30 days. The residual bound water was obtained through the difference between total water content and obtained free and bound water content. In addition, the data in the brackets denoted the percentage content of free water.

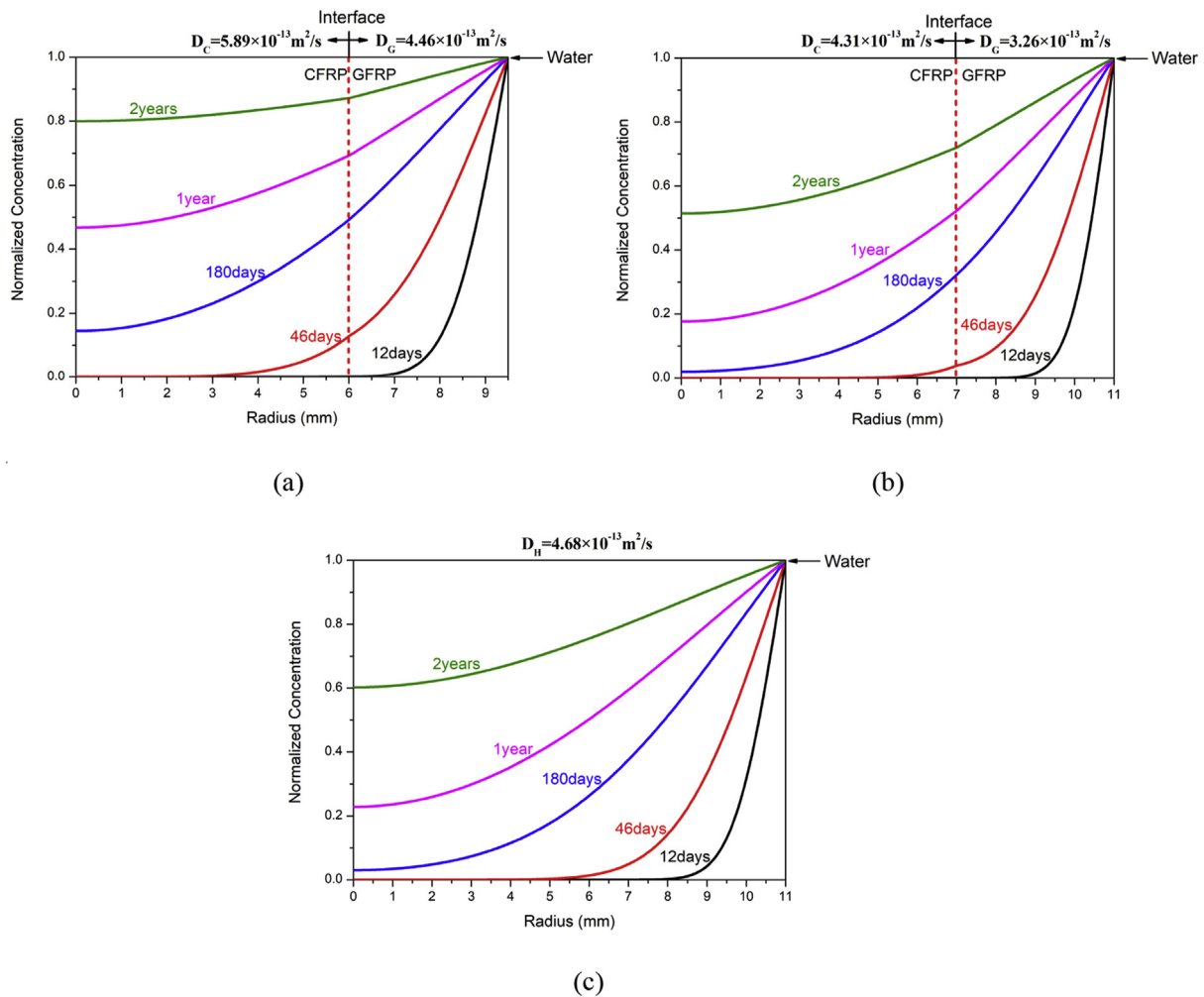


Fig. 5 – Dependence of water concentration distribution on radial position exposed to the freezing-thawing environment of (a) 19-CH, (b) 22-CH and (c) 22-RH.

which indicated the temperature-alternation had the equivalent effect with the constant temperature on the water uptake behavior. For the hydraulic pressure, the increase from 0 to 20 MPa caused an increase of 1.37 times for the diffusivity coefficient. This was attributed to the water molecule were easy to diffuse fast into the microcracks in the hybrid rods under the hydraulic pressure [24]. For the salty concentration, it can be observed that the effects of distilled water and low salty solution concentration on the diffusivity coefficient of hybrid rods were negligible. In comparison, higher salty solution concentration brought about an increase of 1.21 times for the diffusivity coefficient, which was much less than that of immersion temperature, such as 60 and 80 °C/20 MPa (LSSC). For the diameter, the larger diameter increased the interface areas of CFC/GFS and water absorption barrier effect through prolonging the diffusion paths, leading to the lower diffusivity coefficient. Compared to the fiber core–shell hybrid mode (CH), the rods with the fiber random hybrid mode possessed higher diffusivity coefficient, which was attributed

to the different expansion and shrinkage effects between the carbon and glass fibers under the cyclic temperature may cause the formation of microcracks in the CFC/GFS interface and provide more diffusion space for the water molecule. In addition, Table 2 also summarized the diffusivity coefficient of two kinds of fiber core–shell hybrid rods. Obviously, the carbon fiber core had higher diffusivity coefficient compared to the glass fiber shell owing to the larger interfaces of fiber/resin may provide more diffusion paths for the water molecule.

After analyzing the diffusivity coefficients of hybrid rods immersed in different conditions, the maximum water uptakes in several typical service environments (laboratory and actual service environments) were shown in Table 3. As seen, the maximum water uptakes of hybrid rods were mainly dependent on the immersed temperature, and higher temperature brought about a larger water uptake. Through comparing the maximum water uptake of laboratory environment with the actual service environment, it can be observed that the maximum water uptake of hybrid rods had

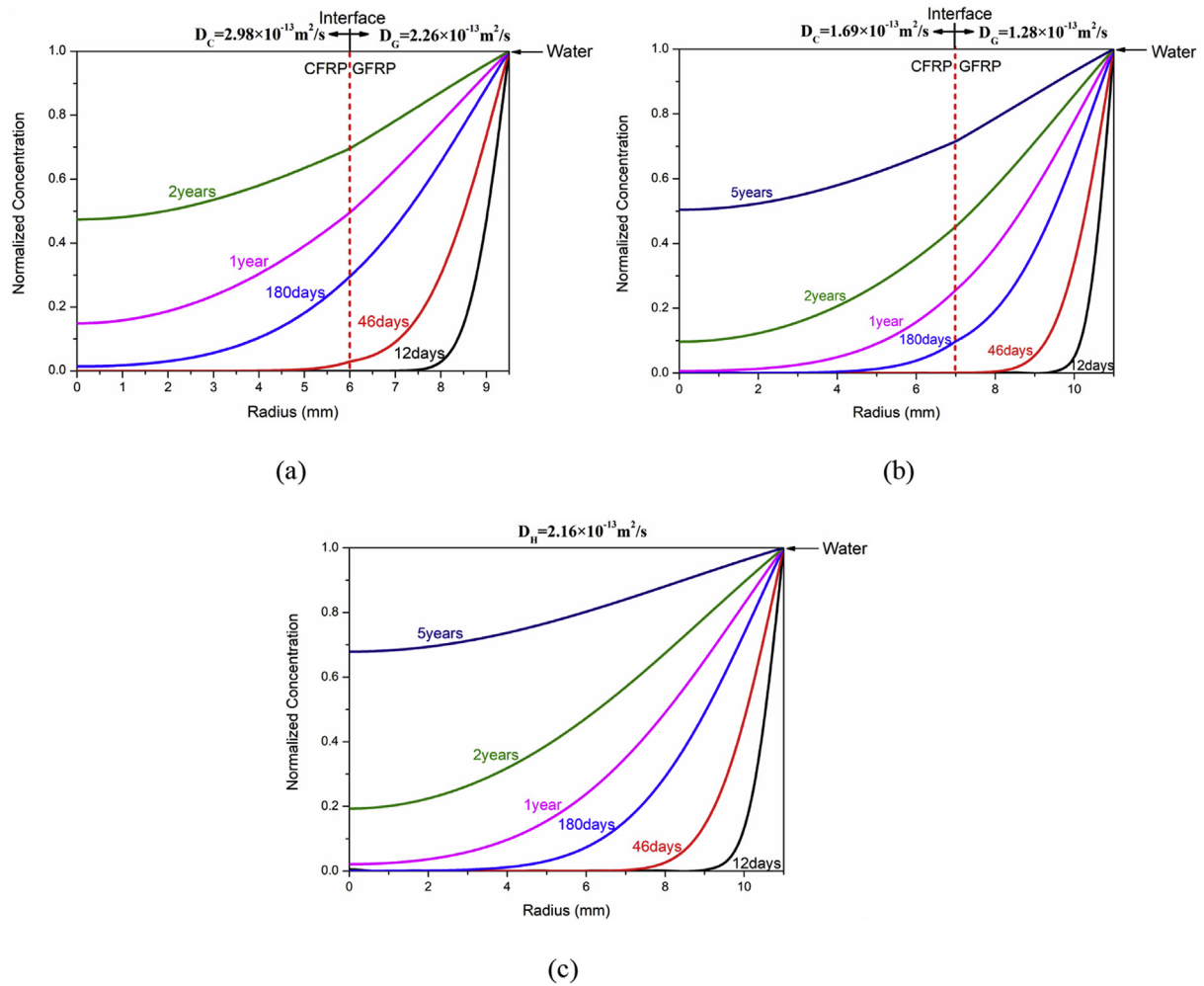


Fig. 6 – Dependence of water concentration distribution on radial position exposed to outdoor environment of (a) 19-CH, (b) 22-CH and (c) 22-RH.

the equivalent effect, such as the maximum water uptake (0.24) of 19-CH exposed in laboratory environment at 20 °C for 365 days was just between 211 (0.198) and 606 days (0.319) exposed in actual service environment at 25 °C. In addition, the fiber random hybrid mode (RH) brought about a slight increase of the maximum water uptake, which was similar to the diffusivity coefficient (Table 2).

To obtain the water molecule type during the immersion, the drying tests of hybrid rods were conducted and the mass losses are shown in Fig. 4. It was noted that the drying temperatures included 60 and 90 °C to obtain the free and bound water molecules [23]. As shown, the mass loss of three kinds of hybrid rods after drying gradually increased and tended to be stable at 60 °C. Based on the stable state, the free water content of hybrid rods during the immersion was obtained and shown in Table 4. It can be seen that the percentage of free water content in the hybrid rods was more than 60%, and achieved the maximum value for 19-CH exposed in FT and OE, which indicated the free water was the main existence form of water molecule in the hybrid rod. The free water attached to the surface of fiber, resin and microcracks. The drying process promoted the fast escape of water molecule from the hybrid

rods. It can be inferred that the effect of free water on the mechanical and thermal properties of hybrid rods was recoverable when they were removed from the rods after drying. Further verification tests on the mechanical and thermal properties of hybrid rods after drying were conducted in the next section. In addition, it can be observed that the mass loss of hybrid rods increased with the drying time at 90 °C. This verified the bound water formed in the hybrid rods during the immersion, and its content increased with the immersed temperature and time. As shown in Table 4, the content of bound water at 90 °C was far more than that of the residual bound water. The effects of bound water on the mechanical and thermal properties of hybrid rods were unrecoverable, because the bound water formed multiple hydrogen bonds with the main and secondary chain structures of resin matrix. Furthermore, a greater amount of accumulated hydrogen bonds possibly brought about the fiber/resin interface debonding, resin plasticization [49] and degradation of interface shear strength.

After obtaining the water uptake of hybrid rods, the effects of exposure environment, rod diameter and fiber hybrid mode on the water concentration distribution were investigated

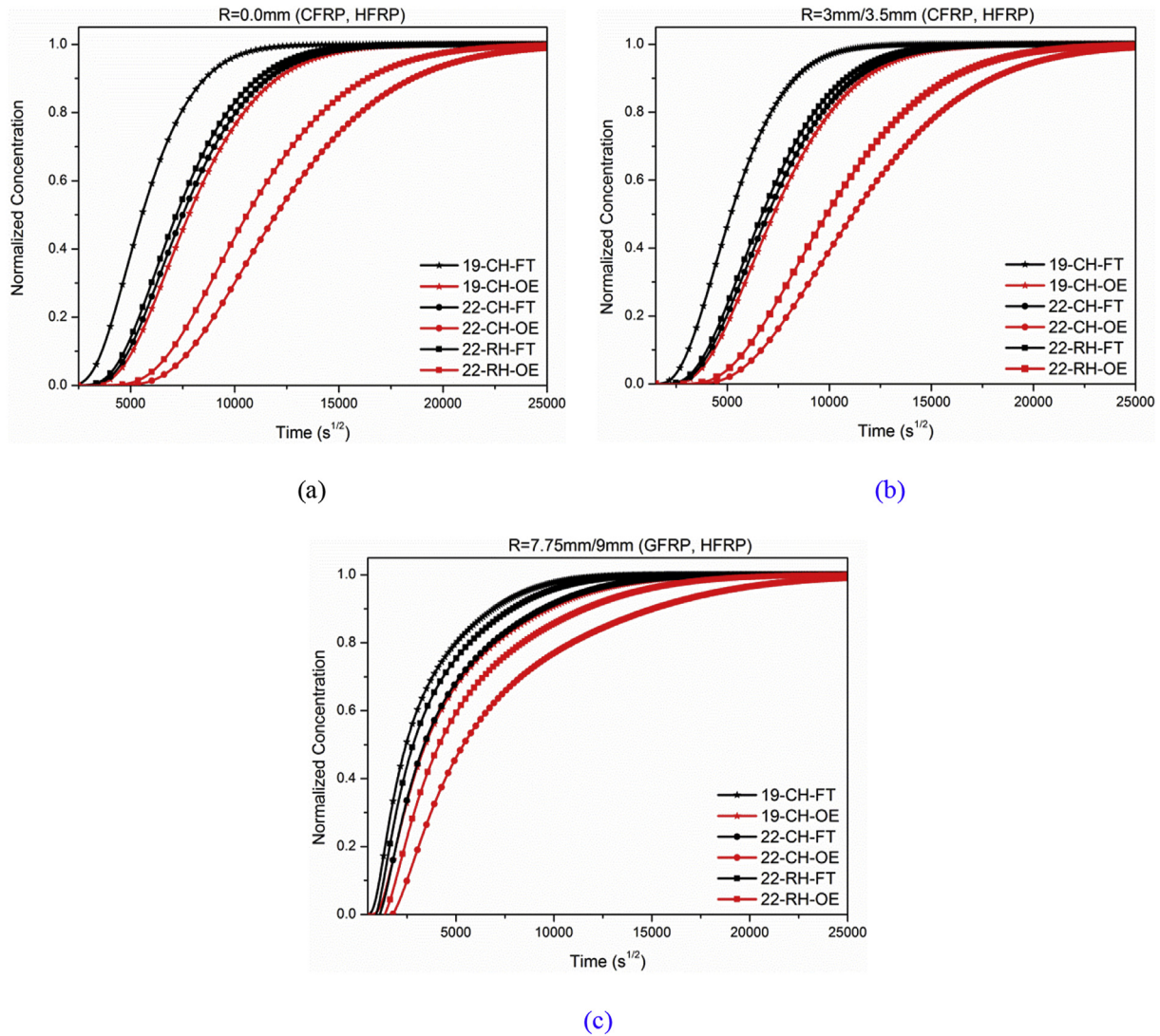


Fig. 7 – Dependence of water concentration radial distribution on the immersion time of (a) $R = 0$ mm (CFRP/HFRP), (b) $R = 3/3.5$ mm (CFRP/HFRP), (c) $R = 7.75/9$ mm (GFRP/HFRP).

through adopting the mathematics diffusion model as shown in Eqs. (3) and (4). Furthermore, the dependences of water concentration distribution on radial position of three kinds of hybrid rods exposed to the freezing-thawing and outdoor environments were shown in Figs. 5 and 6. It should be mentioned that the interface positions of CFC/GFS for 19-CH and 22-CH were different, and they were $R = 6.0$ (R was the radius of hybrid rod) and 7.0 mm, respectively.

$$C_C(r, t) = 1 - \frac{2}{a} \sum_{n=1}^{\infty} \frac{J_0(\alpha_n r)}{J_1(a \alpha_n)} \frac{1}{\alpha_n} \exp(-D_C \alpha_n^2 t), \quad 0 < r < a, \quad (3)$$

$$C_G(r, t) = 1 - 2 \sum_{n=1}^{\infty} \frac{J_0(\alpha_n r)}{b^2 J_1^2(b \alpha_n) - a^2 J_1^2(a \alpha_n)} \frac{1}{\alpha_n} \exp(-D_G \alpha_n^2 t), \quad a < r < b, \quad (4)$$

where $C_C(r, t)$ and $C_G(r, t)$ are the water concentration distributions of CFC and GFS, respectively; a and b are the

radius of CFC and hybrid rod, respectively; J_0 and J_1 are the zero- and first-order Bessel function of the first kind, respectively; and D_C and D_G are the diffusivity coefficients of CFC and GFS, respectively. For the fiber core-shell hybrid mode (19-CH and 22-CH), D_C and D_G were obtained based on the linear combination of diffusivity coefficients in a hybrid rods with a diameter of 7.75 mm determined by Barjasteh et al. [24,45]. For the fiber random hybrid mode (22-RH), the average diffusivity coefficient of hybrid rods was substituted into the Eq. (3) to obtain the water concentration distribution.

As shown in Fig. 5, the water concentration of hybrid rods increased with the radial position from the core to shell layer. Furthermore, the effect of rod diameter on the water concentration was remarkable, such as the water concentrations of core layer ($R = 0.0$ mm) for 2 years were 0.80 (19-CH), 0.51 (22-CH) and 0.60 (22-RH), respectively, which was attributed to the larger diameter increased the diffusion paths of water

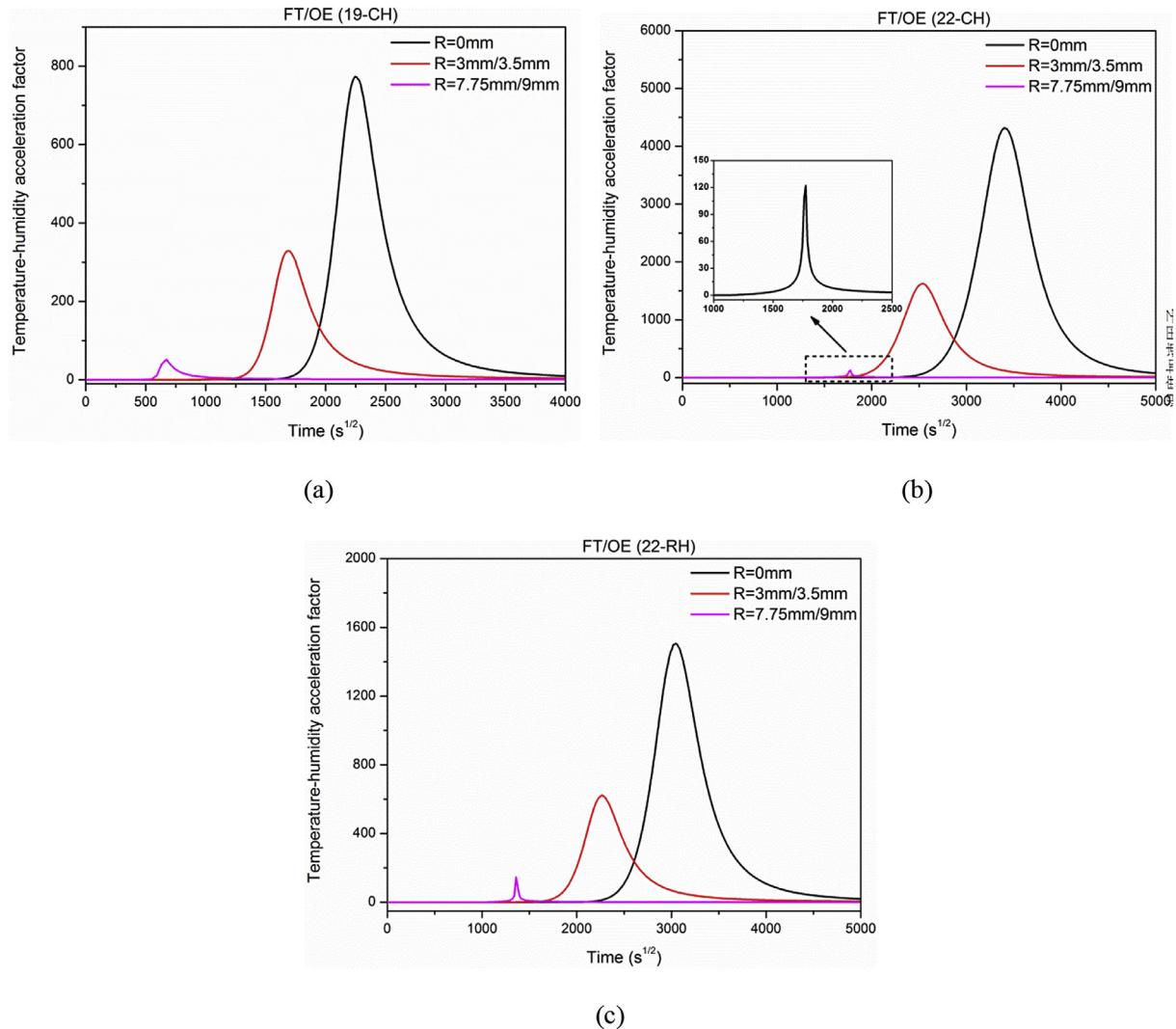


Fig. 8 – Dependence of temperature-alternating acceleration factor on the immersion time of (a) 19-CH, (b) 22-CH and (c) 22-RH.

molecules. By comparison, the effect of fiber hybrid mode on the water concentration was slight.

Fig. 6 shows the water concentration of hybrid rods exposed to the outdoor environment. It can be seen that the water concentrations of hybrid rods were lower than that of freezing-thawing environment. This was because the temperature difference amplitude ($<30\text{ }^{\circ}\text{C}$) and alternating frequency of outdoor environment were less than that of freezing-thawing environment ($65\text{ }^{\circ}\text{C}$). Meanwhile, the maximum exposure temperature of outdoor environment in Harbin was close to $33\text{ }^{\circ}\text{C}$ in summer and the duration was transient. Longer and higher exposure temperature ($40\text{ }^{\circ}\text{C}$) in freezing-thawing chamber accelerated the diffusion of water molecule in the rods, leading to larger water concentration distribution along the radial direction. In addition, it can be observed the effects of rod diameter and fiber hybrid mode on

the water concentration of outdoor environment were similar to the freezing-thawing environment.

After analyzing the radius distribution of water concentration in the hybrid rods, the dependences of normalized water concentration of three typical radial positions on the immersion time were established and shown in Fig. 7. It should be mentioned that the positions of rod center ($R = 0.0\text{ mm}$), core layer ($R = 3.0/3.5\text{ mm}$) and shell layer ($R = 7.75/9.0\text{ mm}$) were selected to obtain the water concentration distribution of three types of hybrid rods. As shown in Fig. 7, the normalized water concentration increased with the immersion time and gradually reached the maximum water absorption. Furthermore, the water concentration curves of hybrid rods exposed in FT were above the curves exposed in OE, which was consistent with the radial distribution of water concentration (Fig. 6).

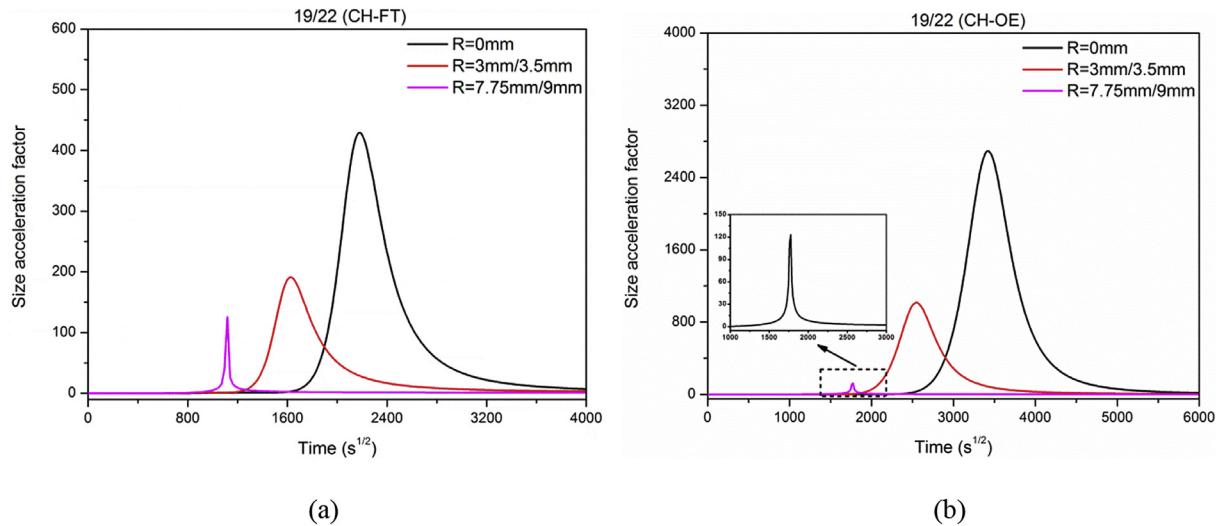


Fig. 9 – Dependence of rod size acceleration factor on the immersion time of (a) CH-FT and (b) CH-OE.

Meanwhile, the effect mechanism of rod diameter and fiber hybrid mode on the water concentration was similar to the above analysis. In addition, it can be observed that the water concentration distribution curves of core layer ($R = 0.0/3.0/3.5\text{ mm}$) with the immersion time were mainly composed of two periods, initial slow increase and subsequent fast increase. In comparison, only fast increase period was observed for the shell layer ($R = 7.75/9.0\text{ mm}$), which was attributed to the delay effect of water absorption from the shell to core layer. In the initial immersion period, the main diffusion path of water molecule was in the shell layer, leading to the lower increase rate of water concentration for the core layer. With the increase of immersion time, the water molecule diffused into the core layer through the CFC/GFS, which brought about the fast increase of water concentration. Furthermore, when the water concentration of rod center was the maximum, the whole rod reached the

water absorption saturation. As a result, the saturated water absorption time was a significant parameter.

To obtain quantitatively the effect of temperature-alternating, rod size and fiber hybrid mode on the water concentration, the acceleration factors of three radial positions through the ratio of water concentration were shown in Fig. 8 (temperature-alternating acceleration factor), Fig. 9 (rod size acceleration factor) and Fig. 10 (fiber hybrid mode acceleration factor). As shown in Fig. 8, the effect of temperature-alternating on the water concentration was remarkable and the maximum temperature-alternating acceleration factors were 773.6 (19-CH), 4316.6 (22-CH) and 1505.5 (22-RH), respectively. The increase of rod size enlarged the difference of water concentration exposed in FT and OE, leading to the increase of acceleration factor. In comparison, the fiber random hybrid mode improved the homogeneity of fiber distribution and weakened the

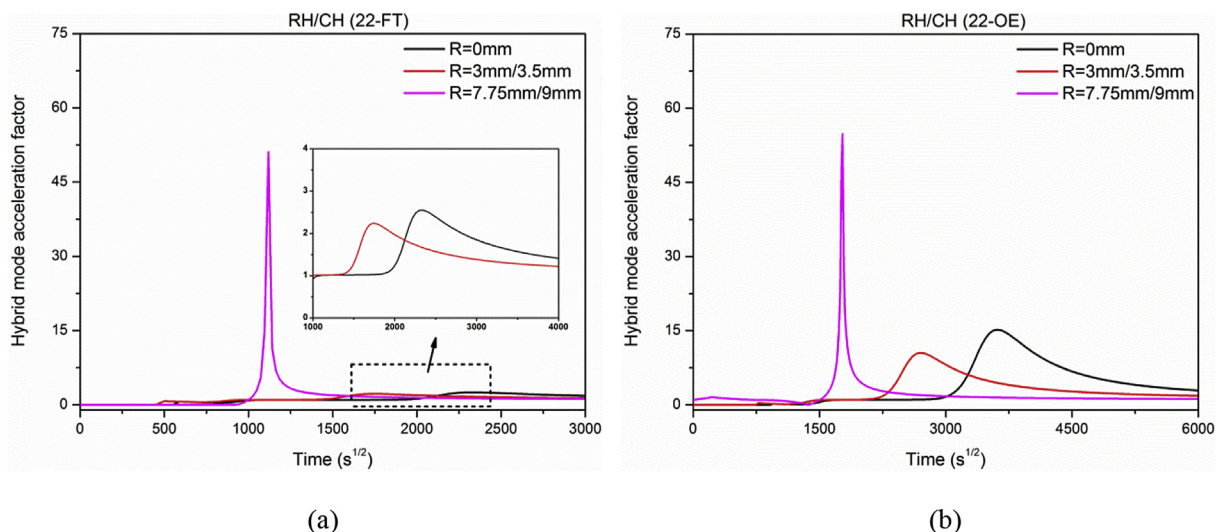


Fig. 10 – Dependence of fiber hybrid mode acceleration factor on the immersion time of (a) 22-FT and (b) 22-OE.

Table 5 – Maximum temperature, temperature-alternating, hydraulic pressure, salty concentration and hybrid mode acceleration factors and acceleration time (days) for $r = 0/3.0/3.5/7.75/9.0$ mm.

Temperature [24,44]	$C_{40}^{\circ C}/C_{20}^{\circ C}$ (DW) 419.9/191.7/21.4 (127/72/12)	$C_{60}^{\circ C}/C_{20}^{\circ C}$ (DW) 17.6E4/5.3E4/559.6 (112/63/11)	$C_{80}^{\circ C-20MPa}/C_{60}^{\circ C-20MPa}$ (LSSC) 38.7/23.2/6.2 (17/10/2)
Temperature-alternating	$C_{FT-19-CH}/C_{OE-19-CH}$ (DW/Air) 773.6/329.7/51.8 (58/33/5)	$C_{FT-22-CH}/C_{OE-22-CH}$ (DW/Air) 4316.6/1620.1/122.1 (134/74/36)	$C_{FT-22-RH}/C_{OE-22-RH}$ (DW/Air) 1505.5/621.7/145.5 (108/60/21)
Hydraulic pressure [24,44]	$C_{60}^{\circ C-20MPa}/C_{60}^{\circ C-0MPa}$ (DW) 30.9/19.2/5.1 (21/12/2)	$C_{60}^{\circ C-20MPa}/C_{60}^{\circ C-0MPa}$ (LSSC) 22.3/14.5/4.8 (23/13/2)	/
Salty concentration [24]	$C_{60}^{\circ C-20MPa}/C_{60}^{\circ C-20MPa}$ (HSSC/LSSC) 8.3/6.2/2.9 (17/10/2)	/	/
Rod diameter	$C_{19-CH-FT}/C_{22-CH-FT}$ (DW) 429.4/191.2/125.9 (55/31/15)	$C_{19-CH-OE}/C_{22-CH-OE}$ (Air) 2693.3/1016.6/123.3 (135/75/36)	/
Fiber hybrid mode	$C_{RH-22-FT}/C_{CH-22-FT}$ (DW) 2.5/2.2/51.1 (63/35/14)	$C_{RH-22-OE}/C_{CH-22-OE}$ (Air) 15.2/10.5/54.8 (151/84/36)	/

Note: C is the water concentration. The data value in the bracket denoted the immersion time when the temperature, temperature-alternating, hydraulic pressure, salty concentration, rod diameter and hybrid mode acceleration factor was maximum.

difference of water concentration, leading to the decrease of acceleration factor.

Fig. 9 shows the dependence of rod size acceleration factor on the immersion time. Compared to the temperature-alternating effect, the rod size acceleration factor was lower. It can be seen that the maximum rod size acceleration factor of hybrid rods exposed in OE was more than that of FT, this was because the higher exposed temperature and temperature-alternating frequency in FT accelerated the diffusion of water molecule in the hybrid rods and reduced the water absorption sensitivity to rod size. Similar explanation was appropriate for the variation of acceleration factor along the radial position. Fig. 10 shows the dependence of fiber hybrid mode acceleration factor on the immersion time. It can be observed that the maximum fiber hybrid mode acceleration factors expose in FT and OE were almost the same, which was far less than temperature-alternating and rod size acceleration factors, indicating the slight acceleration effect of water molecule. In addition, the fiber hybrid mode acceleration factor of shell layer was more than that of core layer, which showed the anomalous trend with the temperature-alternating and rod size acceleration factor. This was because the diffusion coefficient of CFRP was more than that of GFRP, and the random fiber hybrid mode increased the average diffusion coefficient of shell layer ($R = 7.75/9.0$ mm), leading to the increase of acceleration factor. On the contrary, the random fiber hybrid mode reduced the average diffusion coefficient of core layer ($R = 0/3.0/3.5$ mm), leading to the decrease of acceleration factor.

Table 5 summarized the maximum temperature, temperature-alternating, hydraulic pressure, salty

concentration and hybrid mode acceleration factors and acceleration time of hybrid rod exposed at different conditions. It can be seen that the temperature acceleration effect ($C_{60}^{\circ C}/C_{20}^{\circ C}$, 17.6E4) was the most remarkable and reached the maximum value. Furthermore, the acceleration effect was substantially weakened when the immersion temperature increased to 60 and 80 °C. The temperature-alternating acceleration effect was obvious with the increase of rod size. In comparison, the acceleration effect of hydraulic pressure and salty concentration was insignificant. In addition, the acceleration effect of rod size was more than that of fiber hybrid mode. In summary, the larger acceleration factor led to the fast diffusion rate of water uptake, which had significant effect on the degradation of thermal and mechanical properties.

After analyzing the acceleration effect of water uptake for the hybrid rods, the saturated water uptake time was obtained and shown in Table 6 when the normalized water concentration of rod center reached the maximum. As shown, the saturated water uptake time obviously decreased with the increase of immersion temperature, such as it changed from 21.4 (20 °C) to 3.3 years (60 °C). In comparison, the effects of salty concentration, hydraulic pressure and fiber random hybrid mode on the saturated water uptake time were negligible. In addition, it can be observed that the increase of rod diameter brought about the increase percentage of 81.4 (FT) and 136.9% (OE) of the saturated water uptake, indicating the critical accelerating diffusion effect. In conclusion, the maximum water uptake, water concentration, acceleration factors and saturated time of hybrid rod were sensitive to the immersed temperature and rod diameter. The water uptake

Table 6 – Saturated water uptake time of hybrid rod (years).

Temperature [44]	20 °C (DW) 21.4	40 °C (DW) 11.8	60 °C (DW) 3.3	60 °C/20 MPa (DW) 2.4
Salty concentration/Hydraulic pressure [24]	60 °C/0 MPa (LSSC) 3.6	60 °C/20 MPa (LSSC) 2.7	60 °C/20 MPa (HSSC) 2.3	80 °C/20 MPa (LSSC) 1.9
Temperature-alternating/rod diameter/fiber hybrid mode	19-CH-FT 4.3	22-CH-FT 7.8	22-RH-FT 7.2	/
	19-CH-OE 8.4	22-CH-OE 19.9	22-RH-OE 15.6	/

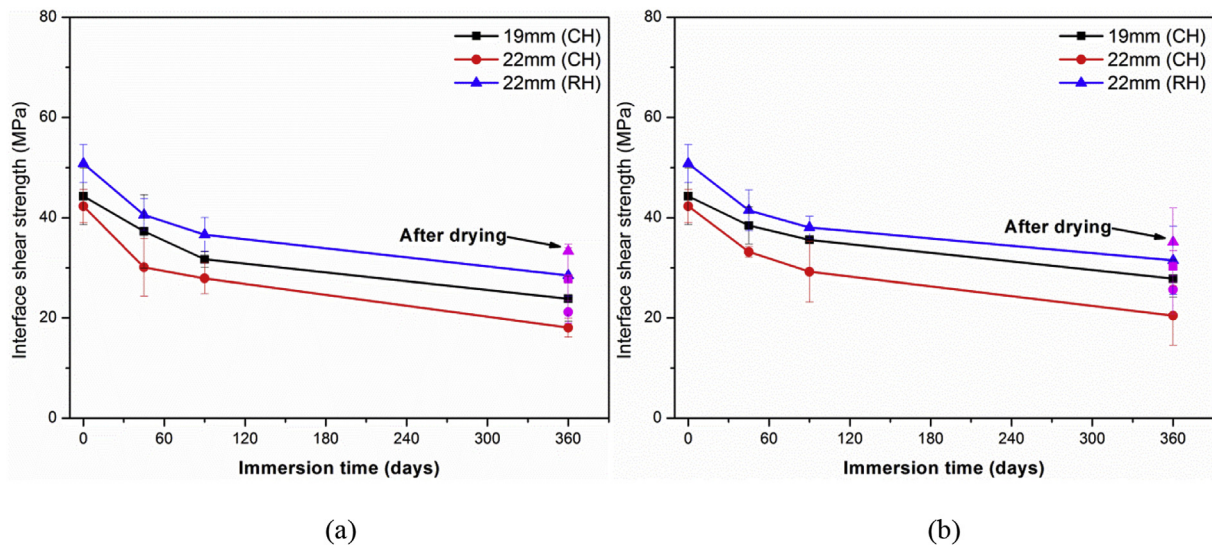


Fig. 11 – Interface shear strength of hybrid rods with the drying time of (a) freezing-thawing environment and (b) outdoor environment.

behaviors of hybrid rods with smaller rod diameter exposed at higher temperatures were remarkable.

3.2. Interface shear strength

Fig. 11 shows the interface shear strength of three kinds of hybrid rods exposed in FT and OE. It can be seen that the obvious degradation of interface shear strength occurred with the immersion time. Maximum degradation percentages were achieved when the immersion time was 360 days, and the strength retentions of three types of hybrid rods were 53.8 (19-CH), 42.7 (22-CH) and 56.0% (22-RH) for the exposure in FT, 62.8 (19-CH), 48.4 (22-CH) and 62.0% (22-RH) for the exposure in OE. Similarly, the lower strength retentions in FT compared to the OE was attributed to the higher immersion temperature and temperature-alternating frequency. In addition, it can be observed that the interface shear strength had the remarkable recovery for the 360 days' immersion samples after drying. This was because the free water of hybrid rods was removed after drying, leading to the increase of interface shear strength. This verified the effects of free water on the mechanical and thermal properties were recoverable. Similar phenomenon was shown in the previous investigation [23]. Furthermore, the removal of free water from the rods brought about the average increase percentage ~16.9% for the three types of hybrid rods in FT and ~15.4% in OE.

To compare the effect of exposure conditions on the interface shear strength, Fig. 12 summarized the variation of interface shear strength with the immersion time exposed at different conditions. Furthermore, the retentions of interface shear strength of hybrid rods after the immersion of 90 days were shown in Table 7. Obviously, the interface shear strength of hybrid rods decreased with the immersion time. The immersion temperature played a decisive role on the degradation of interface shear strength, such as higher temperature

(60 °C) brought about an extra degradation percentage of 17% compared to the immersion temperature at 20 °C. In comparison, the effects of salty concentration and hydraulic pressure on the interface shear strength were negligible and the difference value was less than 5%. The temperature-alternation had a critical effect on the retention of interface shear strength, such as it was 71.6 (19-CH-FT), 77.4 (19-CH-40 °C), 80.3 (19-CH-OE) and 88.5% (19-CH-20 °C), respectively. In addition, it can be observed that the effect of rod size on the interface shear strength was obvious through comparing the retention of 19-CH and 22-CH exposed in FT and OE. This was because the larger diameter increased the formation probability of defects in the rods, such as resin microcracks, microvoids and imperfect bonding of fiber/resin, which provided more diffusion paths and storage space for the water molecule and accelerated the hydrolysis of resin and

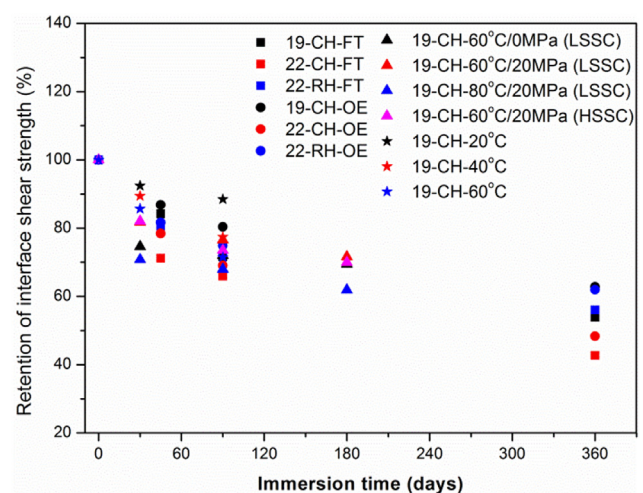


Fig. 12 – Comparison of interface shear strength of hybrid rods exposed at different environments.

Table 7 – Retention of interface shear strength of hybrid rods after the immersion of 90 days (%).

Temperature [44]	19-CH-20 °C (DW)	19-CH-40 °C (DW)	19-CH-60 °C (DW)	/
	88.5	77.4	71.5%	/
Salty concentration/Hydraulic pressure [24]	19-CH-60 °C/0 MPa (LSSC)	19-CH-60 °C/20 MPa (LSSC)	19-CH-60 °C/20 MPa (HSSC)	19-CH-80 °C/20 MPa (LSSC)
	71.9	76.6	73.5	68.0
Temperature-alternating/ rod diameter/fiber hybrid mode	19-CH-FT (DW)	22-CH-FT (DW)	22-RH-FT (DW)	/
	71.6	66.0	72.1	/
	19-CH-OE (Air)	22-CH-OE (Air)	22-RH-OE (Air)	/
	80.3	69.1	74.9	/

debonding of fiber/resin. It should be mentioned that the fiber random hybrid mode had superior durability resistance compared to the fiber core–shell mode, this was because the carbon fiber with smaller diameter filled in the glass fiber with larger diameter, which increased the shear resistance area, leading to the increase of interface shear strength.

3.3. Dynamic mechanical analysis (DMA)

Figs. 13 and 14 show the dependence of glass transition temperature (T_g) on the immersion time exposed in FT and OE. The fiber core–shell structure included the core CFRP and shell GFRP, and the fiber random hybrid structure included HFRP. As seen, T_g of core CFRP and HFRP decreased with the immersion time and the degradation retentions kept almost the same for the hybrid rods exposed in FT and OE. In comparison, the obvious differences of T_g were observed for shell GFRP. T_g retention of fiber random hybrid structure was higher than that of fiber core–shell structure. It can be seen from this that the hybrid rods with the fiber random hybrid structure had higher thermal resistance compared to the fiber core–shell structure. In addition, it can be found that T_g immersed in FT and OT for 360 days had the obvious recovery after drying, which has the similar evolution mechanism with interface shear strength.

3.4. Degradation analysis

The long-term exposure in FT and OE caused the degradation the interface shear strength and T_g. This was because the water molecules from the immersion media entered into the rod and reacted with resins through hydrogen bonding to form the free and bound water, leading to the plasticization and chemical aging of resin, and the deterioration of interfacial bonding [50,51]. To verify the explanation, the dependences of interface shear strength and T_g of hybrid rods on the water uptake were shown in Figs. 15 and 16. As shown in Fig. 15, the linear dependence between the interface shear strength and water uptake indicated the degradation mechanism of interface shear strength owing to water uptake behavior. Furthermore, the linear equations were established to describe the degradation process of interface shear strength. Similarly, the linear expression was also suitable for describing the degradation of T_g, as shown in Fig. 16.

Table 8 shows the recovery of interface shear strength and glass transition temperature of hybrid rods after drying. It should be mentioned that the drying condition was 60 °C for two days. It can be seen that the maximum recovery percentage of interface shear strength was close to 20%. This indicated the removal of free water from the rods after

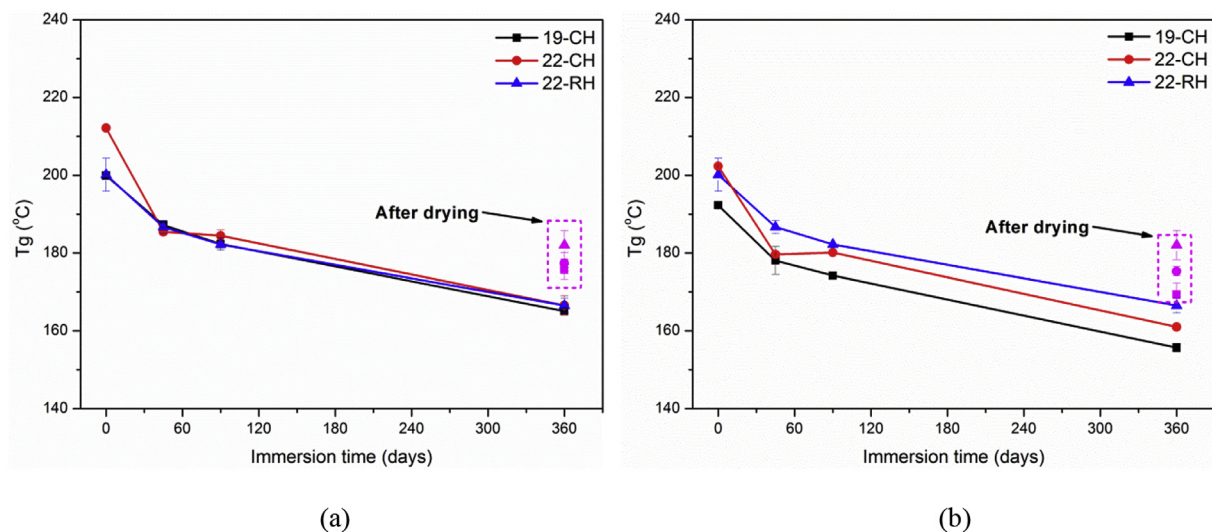


Fig. 13 – Dependence of glass transition temperature on the immersion time exposed in freezing-thawing environments of (a) CFRP/HFRP and (b) GFRP/HFRP.

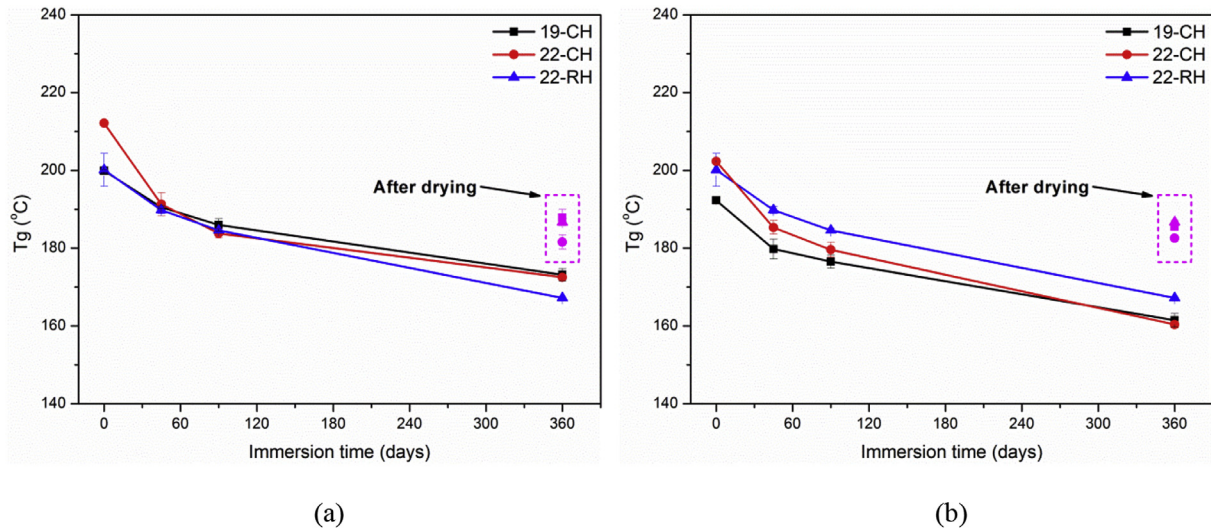


Fig. 14 – Dependence of glass transition temperature on the immersion time exposed in outdoor environment of (a) CFRP/HFRP and (b) GFRP/HFRP.

drying increased of interface shear strength. Similar phenomenon was shown in the recovery of T_g after drying.

Fig. 17 shows the variation of the average density of hybrid rods before and after the immersion. It can be seen that the average density of hybrid rods decreased after the immersion of 360 days, indicating the formation of internal defects. On the one hand, the water molecule diffusion in the rods caused the hydrolysis and plasticization of the resin, forming the microcracks and microvoids [47]. On the other hand, the difference of thermal expansion coefficient among the carbon fiber, glass fiber and resin increased the formation probability of microcracks when exposing in cyclic temperature-alternating. In addition, it can be observed that the density of hybrid rods kept almost unchanged after drying, indicating the irrecoverability of the above degradation process of resin.

3.5. Long-term life prediction

After obtaining the degradation mechanism of interface shear strength, the long-term life prediction of hybrid rods was conducted to provide the design guideline of engineering applications. Based on the above degradation mechanism, the water uptake behavior in the hybrid rod played a significant role on the degradation of interface shear strength. Meanwhile, the linear equations were also established between the water uptake and interface shear strength (Fig. 15). To obtain the long-term life prediction, the relationship between the water uptakes and normalize water concentration of hybrid rods was obtained and shown in Fig. 18. It should be mentioned that normalized water concentrations of shell/core interfaces ($R = 6.0/7.0$ mm) were applied to match the interface shear strength of shell/core interface. It can be seen

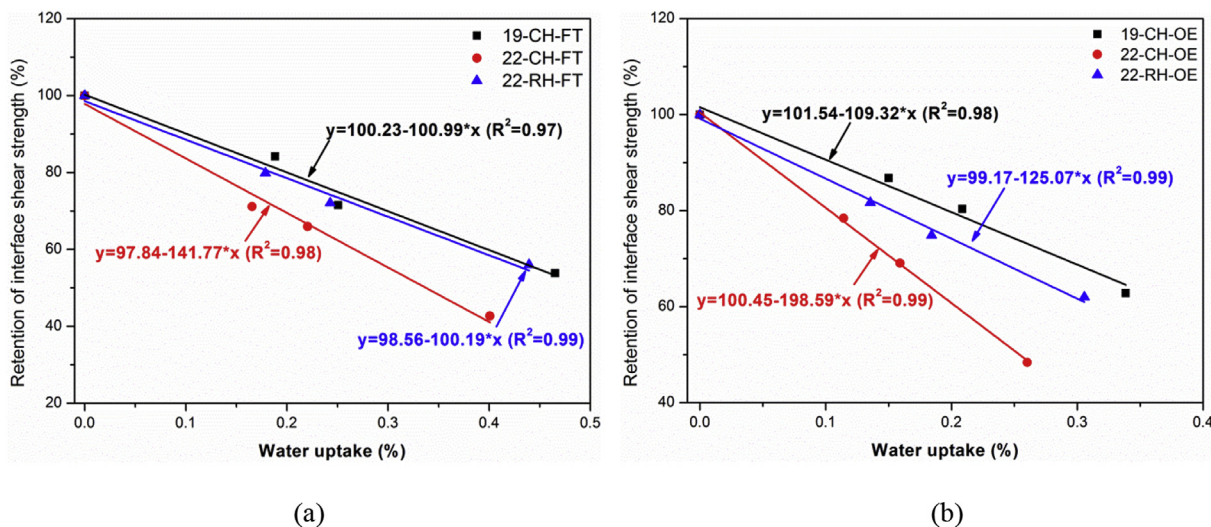


Fig. 15 – Dependence of interface shear strength of hybrid rods on the water uptake of (a) freezing-thawing environment and (b) outdoor environment.

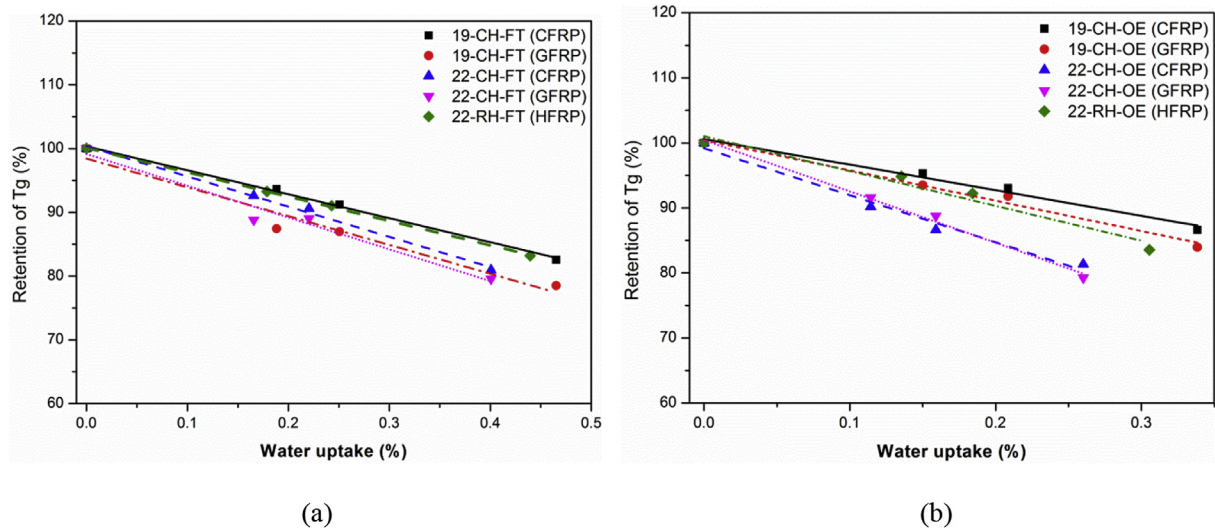


Fig. 16 – Dependence of T_g of hybrid rods on water uptake of (a) freezing-thawing environment and (b) outdoor environment.

Table 8 – Recovery of interface shear strength and glass transition temperature of hybrid rod exposed at two environments for 360 days after drying.

Recovery of interface shear strength (%)	19-CH-FT	22-CH-FT	22-RH-FT
	16.5	17.3	17.0
	19-CH-OE	22-CH-OE	22-RH-OE
	9.0	20.3	11.6
Recovery of glass transition temperature (%)	19-CH-FT	22-CH-FT	22-RH-FT
	6.5/8.8 (CFC/GFS)	6.5/8.9 (CFC/GFS)	9.3 (HFRP)
	19-CH-OE	22-CH-OE	22-RH-OE
	8.5/14.9 (CFC/GFS)	5.2/13.8 (CFC/GFS)	11.7 (HFRP)

that the nonlinear behavior can describe the diffusion process of water molecule from the shell to core for the fiber core-shell structure or from the surface to the center for the fiber random hybrid structure. When referring to the diffusion behavior of water molecule in the hybrid rods (Eq. (2)), the exponential equation can be applied to describe the dependence between the water uptake and normalized water

concentration (Fig. 18). As seen, the higher degree of fitting between the fitting curves and experiment data indicated the rationality of exponential model. Based on the relationship among the interface shear strength, water uptake and normalized water concentration (Figs. 15 and 18), the dependence of interface shear strength on the normalized water concentration was established and shown in Eqs. (5) and (6).

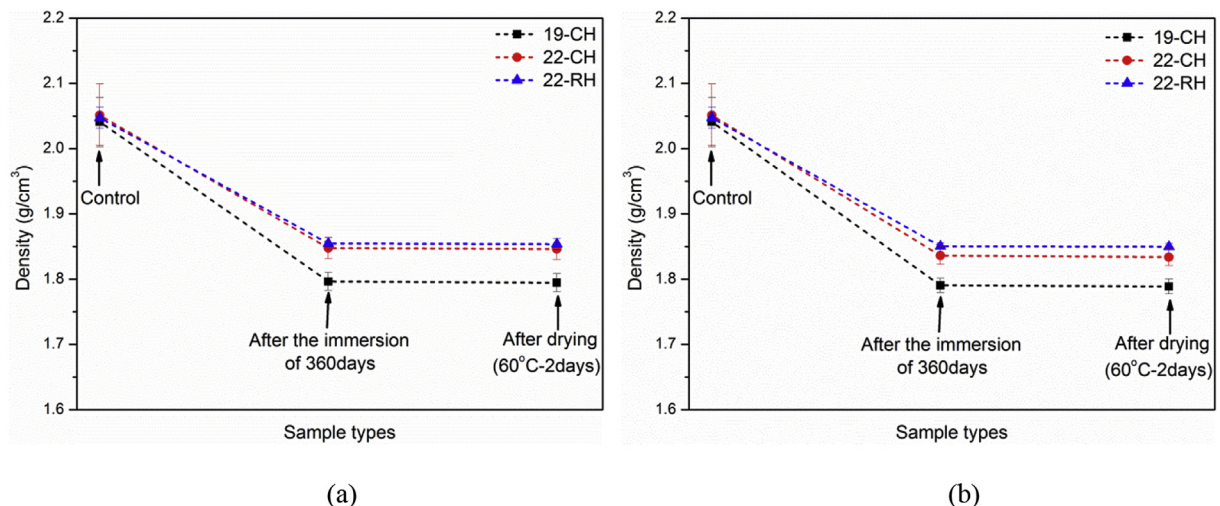


Fig. 17 – Variation of the average density of hybrid rods (a) freezing-thawing environment and (b) outdoor environment.

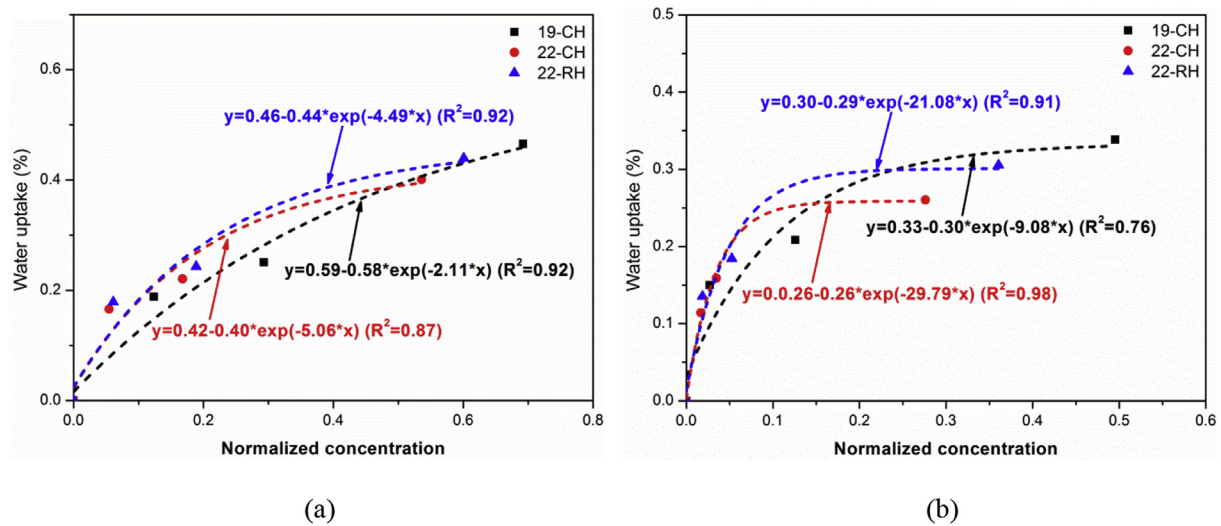


Fig. 18 – Dependence of water uptake of hybrid rod on the normalized concentration (a) freezing-thawing environment and (b) outdoor environment.

$$\begin{aligned}
 \text{ISS}(19 - \text{CH} - \text{FT}) &= 100.23 - 100.99 * (0.59 - 0.58 * \exp(-2.11 * C(r, t))) \\
 \text{ISS}(22 - \text{CH} - \text{FT}) &= 97.84 - 141.77 * (0.42 - 0.40 * \exp(-5.06 * C(r, t))) \\
 \text{ISS}(22 - \text{RH} - \text{FT}) &= 98.56 - 100.19 * (0.46 - 0.44 * \exp(-4.49 * C(r, t))) \\
 \text{ISS}(19 - \text{CH} - \text{OE}) &= 101.54 - 109.32 * (0.33 - 0.30 * \exp(-9.08 * C(r, t))) \\
 \text{ISS}(22 - \text{CH} - \text{OE}) &= 100.45 - 198.59 * (0.26 - 0.26 * \exp(-29.79 * C(r, t))) \\
 \text{ISS}(22 - \text{RH} - \text{OE}) &= 99.17 - 125.07 * (0.30 - 0.29 * \exp(-21.08 * C(r, t)))
 \end{aligned}
 \tag{5}$$

where ISS is interface shear strength and $C(r, t)$ is the normalized water concentration of shell/core interfaces.

Based on Eqs. (3)–(6), the long-term life predictions of interface shear strength for three kinds of hybrid rods exposed in FT and OE were finally obtained and shown in Fig. 19. As shown, the retention of interface shear strength presented on the initial decrease and subsequent stable condition. The decrease process of interface shear strength was owing to the water uptake and diffusion and the subsequent stable condition was realized when the saturated water absorption was completed. Table 9 also shows the stable retention of interface shear strength of hybrid rods exposed at different conditions. It can be seen that the temperature-alternating contributed to an extra degradation up to 31.2% of stable strength retention

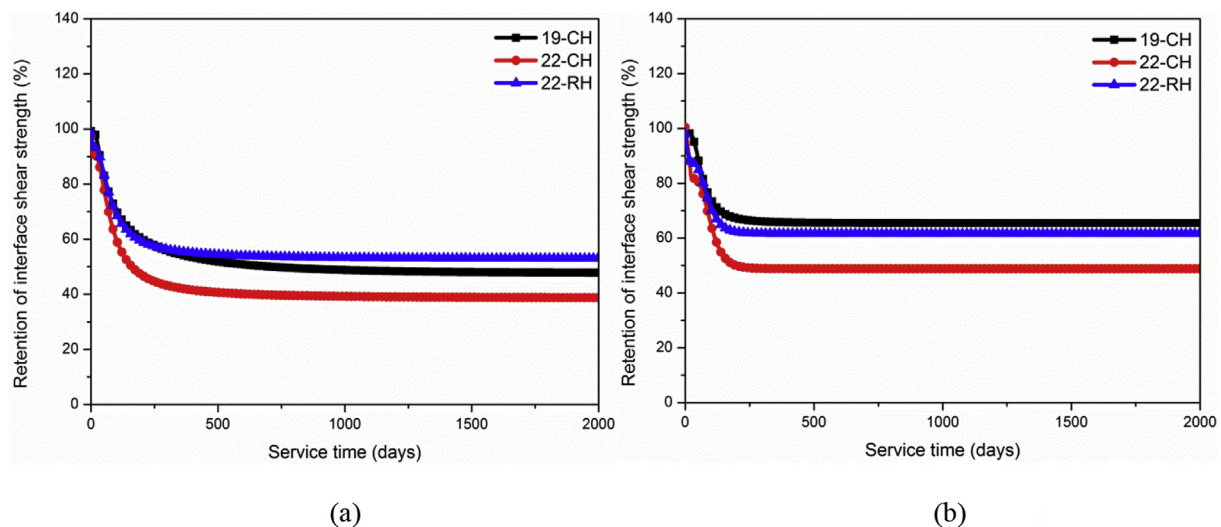


Fig. 19 – Life prediction of interface shear strength of hybrid rods of (a) freezing-thawing environment and (b) outdoor environment.

Table 9 – Stable retention of interface shear strength of hybrid rods exposed at different conditions (%).

Temperature [6]	19-CH-20 °C (DW) 69.9	19-CH-40 °C (DW)	19-CH-60 °C (DW)
Salty concentration/hydraulic pressure [24]	19-CH-60 °C/20 MPa (LSSC) 54.18	19-CH-60 °C/20 MPa (HSSC)	19-CH-80 °C/20 MPa (LSSC)
Temperature-alternating/rod diameter/fiber hybrid mode	19-CH-FT 47.8 19-CH-OE 65.5	22-CH-FT 38.7 22-CH-OE 48.8	22-RH-FT 53.0 22-RH-OE 61.6

compared to the constant temperature. Furthermore, the salty concentration and hydraulic pressure also accelerated the degradation of interface shear strength. The increase of rod size obviously decreased the stable retention of interface shear strength owing to the formation probability increase of initial defects. In addition, the rod with fiber random hybrid structure possessed the superior resistance to FT and OE compared to the core–shell structure, leading to the increase percentage up to 14.3%.

4. Conclusions

In the present paper, three kinds of carbon/glass fiber reinforced polymer composite rods were developed. Effects of rod diameter and fiber hybrid mode on the mechanical properties exposed to the laboratorial freezing-thawing and actual outdoor environments were investigated.

- (1) Fick law can describe the diffusion behavior of three kinds of hybrid rods exposed in FT and OE. The fiber random hybrid mode brought about a slight increase of water uptake.
- (2) The effects of temperature and temperature-alternating on acceleration factors and saturated water uptake time were remarkable compared to the hydraulic pressure and salty concentration. Furthermore, the increase of rod diameter increased the saturated water uptake time of 81.4 (FT) and 136.9% (OE).
- (3) Immersed in FT led to the obvious degradation of interface shear strength compared to OE owing to the hydrolysis and plasticization of the resin. In addition, the remarkable recovery up to 20.3% of interface shear strength after drying occurred owing to the removal of free water.
- (4) Temperature-alternating contributed to an extra degradation percentage up to 31.2% of stable strength retention compared to the constant temperature. In addition, the rods with the fiber random hybrid mode and smaller diameter possessed the superior corrosive resistance.

Funding

This research work was funded by National Natural Science Foundation of China (52008137 and 51908165), China Post-doctoral Science Foundation (2019TQ0079 and 2019M661288) and Heilongjiang Province Natural Science Foundation (LH2021E073).

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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